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Explainable AI with SHAP and Grad-CAM: A Novel Approach to Visualizing Cassava Leaf Health using CIVE, ECI Indices, and an Advanced UNet Framework

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***Abstract*— This paper presents an in-depth study on the classification and segmentation of both healthy and diseased Cassava leaves by visualizing traditional image processing techniques with deep learning approaches, and introducing explainability to enhance model transparency. Agricultural monitoring typically relies on expensive multispectral sensors to assess plant health, but this research explores the use of affordable RGB cameras. Two vegetation indices, the Elliptical Color Index (ECI) and Color Index for Vegetation Extraction (CIVE), are applied to extract vegetation features from RGB images, with Otsu thresholding used to create binary masks for segmentation. While CIVE consistently outperforms ECI in segmenting leaf regions, both methods struggle to distinguish between healthy and diseased leaves effectively. To address these limitations, a UNet deep learning model was trained on the binary masks generated by the indices, achieving significantly improved segmentation accuracy through techniques like data augmentation and optimized hyperparameters. UNet outperformed traditional methods, reaching 97.36% accuracy, 93.14% precision, 94.27% recall, an F1-score of 96.11%, an IoU of 91.08%, and a Dice coefficient of 92.32%. A novel aspect of this study is the integration and comparison of Explainable AI (XAI) techniques, including Grad-CAM and SHAP, to provide visual insights into how the UNet model distinguishes between healthy and diseased regions. These tools offer a deeper understanding of the model's decision-making process, with SHAP proving particularly effective in highlighting critical areas of focus. This comprehensive evaluation of traditional image processing, deep learning, and XAI methods contributes to a more accurate and comprehensive solution for plant health monitoring, addressing both performance and transparency challenges in agricultural applications.**

Keywords*— *Cassava Leaf Classification, ECI (Enhanced Color Index), CIVE (Color Index for Vegetation Extraction), UNet, SHAP, Grad-CAM

I. Introduction

In the realm of remote sensing, identifying plant health from images is critical for monitoring agricultural conditions, particularly for staple crops like Cassava (Manihot esculenta). Cassava is a vital food source in tropical regions, providing sustenance and income for millions of people. However, diseases affecting Cassava plants can lead to yield losses, making early detection essential. Remote sensing technologies, particularly multispectral and hyperspectral imaging, have emerged as vital tools for evaluating plant health by detecting a broad range of spectral bands beyond the visible spectrum. These methods leverage indices such as the Normalized Difference Vegetation Index (NDVI) to distinguish healthy vegetation from stressed or diseased plants. However, the high costs and complexities associated with these sensors often limit their accessibility, particularly for smallholder farmers in developing regions.

Traditional image processing techniques, which are more cost effective and widely available, have prompted the exploration of methods utilizing RGB cameras. Vegetation indices like the Elliptical Colour Index (ECI) and the Colour Index for Vegetation Extraction (CIVE) have been proposed to extract and analyse vegetation from RGB images. These indices offer a viable solution for assessing plant health without the need for expensive multispectral sensors, making them suitable for broader applications. However, while ECI and CIVE can effectively segment healthy plants, their performance in classifying diseased leaves remains suboptimal, underscoring the necessity for improved methodologies.

Deep learning, particularly convolutional neural networks (CNNs), has demonstarted remarkable promise in image segmentation and classification tasks. The UNet architecture, designed specifically for biomedical image segmentation, has gained popularity for its ability to accurately delineate structures within images. By leveraging the power of deep learning, it is possible to enhance the segmentation and classification of both healthy and diseased Cassava leaves, which are a staple crop in many tropical regions. Despite the advancements in deep learning, a strong challenge persists regarding the need for transparency and interpretability in model predictions. As these

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models become increasingly complex, understanding their decision-making processes has become essential, especially in critical fields such as agriculture. This study aims to bridge the gap between traditional image processing techniques and modern deep learning approaches by comparing their effectiveness in classifying Cassava leaf health while incorporating Explainable AI (XAI) to enhance interpretability.

The objective of this research is to improve the segmentation and classification of both healthy and diseased Cassava leaves using deep learning while maintaining transparency and interpretability through Explainable AI (XAI). To achieve this objective, the study will evaluate the performance of ECI and CIVE vegetation indices combined with Otsu thresholding, a technique that determines an optimal threshold by minimizing intra-class variance between pixel intensity values to enhance the leaf segmentation, train a UNet model for image segmentation, and apply XAI techniques like SHAP and Grad-CAM and also compare them to elucidate the model's decision-making process. Through this detailed analysis, valuable insights will be provided regarding the efficacy of traditional versus modern methodologies, and also between various XAI algorithms in the context of agricultural health monitoring.

The subsequent sections of the paper include the Literature Review, Proposed Methodology, Results and Discussion, as well as the Conclusion and Future Work.

II. Literature survey

In this section, we conduct a comprehensive review of the existing literature related to Cassava Leaf Health Classification using explainable AI Techniques. By synthesizing these findings, we aim to provide a contextual foundation for the proposed methodology and underscore the necessity for continued innovation in this domain.

Recent studies in image processing and deep learning have increasingly focused on leveraging advanced techniques for effective monitoring of plant health, particularly in agricultural settings. Prashanth et al [1] presents recent advancements in augmented reality (AR) and deep learning techniques to enhance Cassava leaf disease detection, crucial for improving agricultural productivity in Sub-Saharan Africa and Southeast Asia. The Modified Pyramidal Convolutional Shuffle Attention Residual (MPCSAR-AHH) model achieves an impressive accuracy of 99.02%, alongside 97.55% precision and recall. Future developments may leverage drones with high resolution cameras for efficient and large scale monitoring of Cassava fields.

Zhang et al [2] presents a novel multi-head attention IBN antialiasing neural network (MAIANet) designed to improve Cassava leaf disease classification accuracy, addressing the threats posed by viral and bacterial diseases to food security. The MAIANet employs advanced techniques, including a multi-head attention method for optimal feature fitting and an anti-aliasing block for enhanced high-frequency feature extraction. With an accuracy of 95.83%, a loss of 1.720, and an F1-score of 0.9585, the proposed model significantly outperforms traditional networks like V2-ResNet-101 and EfficientNet-B5. Haider et al [3] presents two novel architectures—Efficient Feature Fusion Segmentation Network (EFFS-Net) and Multi-Scale Dilated Feature Fusion Segmentation Network (MDFS-Net)—to automate the segmentation of plant diseases and fruits in agricultural images. EFFS-Net employs an identity skip connection-based feature fusion mechanism, while MDFS-Net enhances segmentation performance using multi-scale feature fusion and multiple receptive fields. The models achieve impressive Dice similarity coefficients of 88.81%, 95.01%, and 86.04% on various datasets, demonstrating high computational efficiency with only 2.1 million trainable parameters. This research addresses challenges in existing AI-based methods, paving the way for more precise agricultural practices.

Raja et al [4] presents a Radial Function Neural Network optimized with a SalpSwarm Algorithm (PLDC-RBFNN-SSA) for the segmentation and classification of rice leaf diseases, addressing huge yield losses of 20–40% in rice production. The methodology includes image preprocessing through augmentation techniques and segmentation using k-means clustering, followed by feature extraction via an adaptive grayscale co-occurrence matrix. The proposed approach achieves substantial performance improvements, with sensitivity, specificity, precision, and F1-scores enhanced by up to 26.75%. The, PLDC-RBFNN-SSA demonstrates a 50.2% reduction in computation time compared to existing methods, highlighting its efficiency and accuracy in real world applications amidst challenges such as lighting discrepancies and background variations.

Sharma et al [5] introduces a novel Multi-Task Joint Learning Network (MTJNet) designed to automate the classification of medicinal plant leaves, addressing the inefficiencies of manual methods that require expert knowledge. By integrating a comprehensive dataset of medicinal plants and employing preprocessing techniques, such as leaf vein morphometric features and Sobel edge detection, MTJNet effectively highlights key image characteristics. The network achieves impressive results with a precision of 99.60%, recall of 99.62%, accuracy of 99.71%, and F1 score of 99.58%.

Biswas et al [6] addresses the critical issue of agricultural yield loss due to pest attacks and diseases, proposing a new energy-efficient convolutional neural network (CNN) architecture for plant disease detection. The model is evaluated against established architectures, VGG19 and Inception V3, using datasets from Plant Village, Cassava, and rice crops. The proposed CNN achieves impressive accuracy rates of 95.17% for the Plant Village dataset, 99.8% for rice, and 63% for Cassava, outperforming VGG19 and Inception V3 on the first two datasets while demonstrating comparable performance on the Cassava dataset.

Demilie et al [7] elaborates the critical need for efficient detection and classification of plant diseases to mitigate agricultural losses with various deep learning (DL) techniques, particularly convolutional neural networks (CNNs), as traditional methods are laborious and prone to error. The study emphasizes the potential for improved accuracy through the use of extensive, diverse datasets and advanced data augmentation techniques, while also recommending the development of real-

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time detection systems and mobile applications for practical implementation

Owomugisha et al [8] elaborates on the matrix relevance learning for diagnosing Cassava diseases using spectral data, offering a novel approach to detecting non symptomatic infections. The research compares multiple algorithms, including Generalized Matrix Relevance Learning Vector Quantization (GMLVQ), Support Vector Classifier (SVC), and Convolutional Neural Networks (CNN), achieving maximum accuracy with the SVC. Challenges include managing the high dimensionality of spectral data and ensuring effective feature reduction for CNN performance.

Feng et al [9] addresses the limitations of traditional crop disease detection methods by proposing the EfficientNet Convolutional Group-Wise Transformer (EGWT) model, which leverages EfficientNet for feature extraction and a group-wise transformer for attention-based feature correlation. The model achieves high accuracy—99.8% on the PlantVillage dataset, 86.9% on the Cassava dataset, and 99.4% on the Tomato leaves dataset. Core issues include handling imbalanced datasets and maintaining computational efficiency in real-world agricultural applications. Future work of studies will focus on extending models to agricultural tasks like weed detection [10] and ensuring agricultural sustainability for farmers [11].

III. Proposed Methodology

This study presents a comprehensive methodology that visualizes traditional vegetation indices with advanced deep learning models and integrates novel Explainable AI (XAI) algorithms to enhance the segmentation and classification of healthy and diseased Cassava leaves. Figure 1 represents both healthy and diseased samples from the training set. The dataset of healthy and diseased Cassava leaves comprises of approximately 5,000 training images and 1,000 testing images, maintaining a ratio of 5:1 as shown through the proposed framework in Figure 2. The methodology is designed to address the limitations of conventional approaches [12]. Traditional vegetation indices, such as the Elliptical Colour Index (ECI) and the Colour Index for Vegetation Extraction (CIVE), serve as foundational tools for analysing plant health based on visual characteristics. While these indices provide valuable insights, they often fall short in accurately identifying subtle variations in leaf conditions that can indicate the onset of vegetation or disease. To bridge this gap, the methodology incorporates the UNet deep learning architecture, which has proven highly effective in various image segmentation tasks due to its ability to capture intricate patterns and features within images.



Fig. 1. Sample images from the dataset

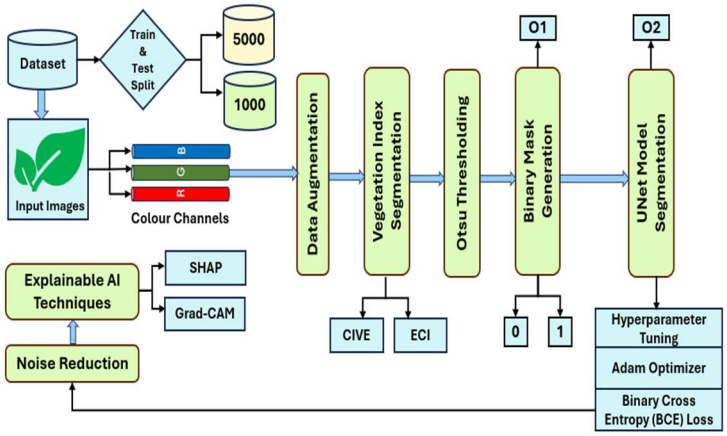


Fig. 2. Proposed Methodology Framework

By leveraging deep learning, the proposed methodology seeks to improve the accuracy and robustness of segmentation outcomes, even in challenging conditions characterized by noise or varying lighting. Furthermore, the incorporation of Explainable AI techniques, particularly Grad-CAM and SHAP, enhances the interpretability of model decisions, providing valuable insights into the factors influencing predictions. This comparison of traditional and advanced methodologies allows for a more holistic understanding of Cassava leaf health, enabling farmers and agricultural professionals to make informed decisions based on reliable data. Ultimately, this approach aims to facilitate timely interventions, thereby improving crop yield and sustainability in Cassava production, which is a staple food worldwide and accounts for approximately 30% of the daily caloric intake for over 800 million people globally.

A. Comparison of Vegetation Indices: ECI and CIVE for Effective Segmentation of Cassava Plants

In order to obtain efficient and accurate vegetation analysis, this section focuses on the testing and comparison of two key vegetation indices: the Elliptical Colour Index (ECI) and the Colour Index for Vegetation Extraction (CIVE). The primary objective is to utilize these indices in conjunction with Otsu thresholding to extract vegetation from RGB images of Cassava plants. The publicly available dataset comprises a diverse collection of images, encompassing both healthy and diseased Cassava leaf samples, facilitating a robust analysis of the indices' effectiveness in real-world scenarios. The process begins with the preparation of input data, which includes RGB images capturing various states of Cassava foliage. To highlight the vegetative features within these images, ECI and CIVE indices were implemented and computed. These indices are instrumental in emphasizing the specific colour characteristics associated with healthy vegetation. Once the masks are generated, Otsu thresholding is applied to their outputs, transforming the continuous grayscale representations into binary masks. This critical step allows for the clear differentiation between vegetation and non-vegetation areas within the images.

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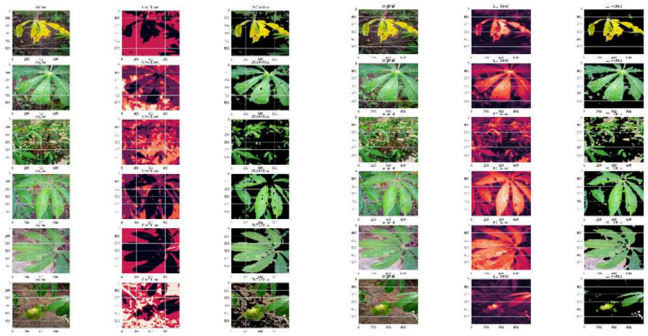


Fig. 3. RGB image, band, and output with Otsu Thresholding for (a) ECI (b) CIVE

The evaluation of the binary masks derived from the ECI and CIVE indices forms a crucial part of this analysis. A visual assessment was conducted to envision the performance of the ECI combined with Otsu thresholding against that of the CIVE combined with Otsu thresholding. The output consists of three columns: the first represents the original RGB image, the second depicts the ECI band, which enhances the visibility of vegetation by filtering the background from the foreground, and the third illustrates the binary mask produced by the ECI combined with Otsu thresholding, as shown in Figure 3 (a). For CIVE, the output also includes a similar format as shown in Figure 3 (b): the original RGB image, the CIVE band, which specifically focuses on colour features to differentiate vegetation from non-vegetation, and the binary mask generated through the application of Otsu thresholding.

Figure 4 showcases ECI with Otsu and CIVE with Otsu combination outputs side by side, facilitating a direct visual comparison. Through this comprehensive analysis, the superiority of the CIVE index over the ECI index in effectively segmenting vegetative regions is demonstrated.

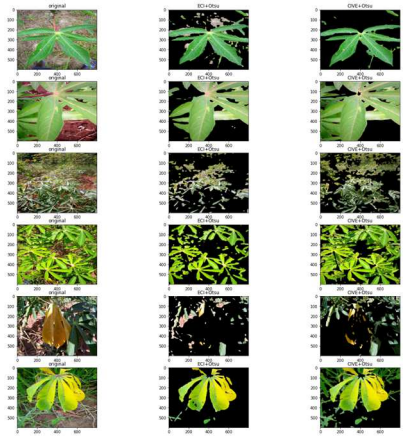


Fig. 4. Comparison of ECI and CIVE Indices for Classifying Cassava Leaves

B. Training a Deep Learning Model (UNet) for Cassava Leaf Segmentation Analysis

This section describes the implementation of the proposed UNet model, a powerful fully convolutional neural network renowned for its efficacy in image segmentation tasks. This approach is particularly beneficial for the segmentation of Cassava leaves, enabling the distinction between healthy and diseased plants. To enhance the model's performance and robustness, a comprehensive data augmentation strategy is employed. This includes techniques such as rotation (±40 degrees), mirror imaging, brightness adjustment (to 0.7), contrast enhancement (to 1.2), grayscale conversion, and spatial shifting (up, down, left, and right by 0.4).

The training process utilizes the Adam optimizer, chosen for its adaptive learning rate capabilities, and is conducted over 13 epochs. During this phase, hyperparameters are meticulously tuned, including a learning rate of 0.0000001, a batch size of 64, 128 filters in the convolutional layers, and a dropout rate of 0.3. These settings are designed to optimize the model's learning capacity while minimizing the risk of overfitting. Once the model is trained, it is tasked with segmenting the Cassava leaf images, producing output that distinctly separates plant regions (vegetation) from the background. The efficacy of the UNet model is then assessed through both visual assessments and quantitative metrics.

To visualize the performance of the UNet model with traditional vegetation indices, the segmentation results generated by the UNet against those derived from the ECI and CIVE methods are evaluated, as shown in Figure 5

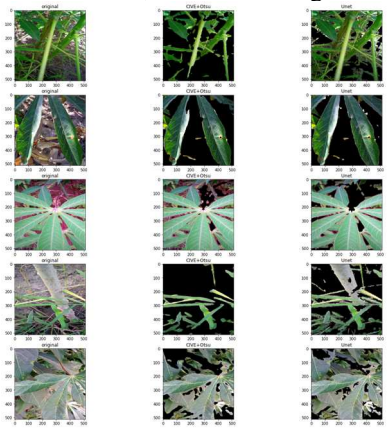


Fig. 5. Comparison of the outputs produced by the UNet deep learning model and the traditional indices (ECI and CIVE)

C. Enhancing Visualization through Explainable AI: Comparing Grad-CAM and SHAP for Interpretability

In our study, a novel approach to enhancing model transparency is introduced by integrating Explainable AI (XAI) techniques, specifically Gradient-weighted Class Activation Mapping (Grad-CAM) and SHapley Additive exPlanations (SHAP). The analysis allows us to visualize the regions of these thereby assessing its ability to accurately identify diseased regions in Cassava leaves. It also highlights how the model's decision-making diverges from traditional vegetation indices. To begin our examination of XAI, Grad-CAM and SHAP are applied to a sample noise image processed by the UNet model as shown in Figure 6.leaf images that the UNet model prioritizes during segmentation, Grad-CAM produces heatmaps that indicate parts of an image responsible for most model prediction, whereas SHAP actually assigns to every pixel a contribution to the final output, so it can give more complex

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insights about the model's behavior. When applied to the sample noise image, the heatmaps generated by both methods reveal distinct focus areas: Grad-CAM highlights regions in blue, signifying less importance, while SHAP displays a broader range of colour intensities, particularly light blue, grey, and greenish areas, indicating higher importance across a more extensive area of the image.

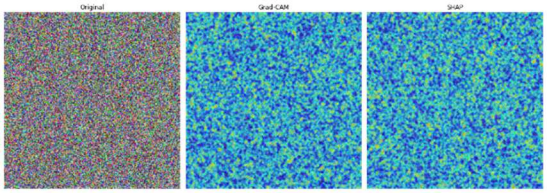


Fig 6. Grad-CAM and SHAP heatmaps applied to a sample noisy image processed by the UNet model for interpretability.

Explainability in Cassava leaf segmentation allows for precise identification of diseased areas, enabling targeted interventions and reducing the risk of crop loss by empowering farmers to make data-driven decisions based on model insights. A comparative analysis is also performed between the two algorithms, aiming to elucidate how each method interprets model predictions and identify their effectiveness in isolating diseased patches in Cassava leaves. Therefore, the integration of these techniques represents an advancement in understanding deep learning model decisions in the context of agricultural image segmentation, contributing novel insights to the field.

IV. Results and Discussions

The performance of the UNet model was evaluated using six critical metrics: accuracy, precision, recall, F1-score, Intersection over Union (IoU), and Dice coefficient, over a span of 13 epochs. The results indicate that the UNet model achieved the highest accuracy of 97.36%, with corresponding precision of 93.14%, recall of 94.27%, F1-score of 96.11%, IoU of 91.08%, and Dice values of 92.32%. In comparison, alternative models such as FCN, DeepLabV3, SegNet, and PSPNet yielded accuracies of 91.12%, 92.21%, 89.35%, and 91.02%, respectively. Further analysis of model performance is illustrated in Figure 7 (a) and (b), which presents the accuracy and loss trends during both training and validation phases. Notably, the UNet model exhibits a consistent increase in accuracy alongside a gradual reduction in loss over the 13 training and validation epochs, indicating successful learning and robust differentiation between healthy and diseased leaves without signs of overfitting.

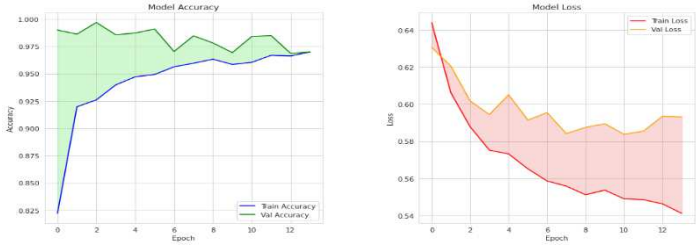


Fig. 7. Training and Validation (a) Accuracy curves (b) Loss curves

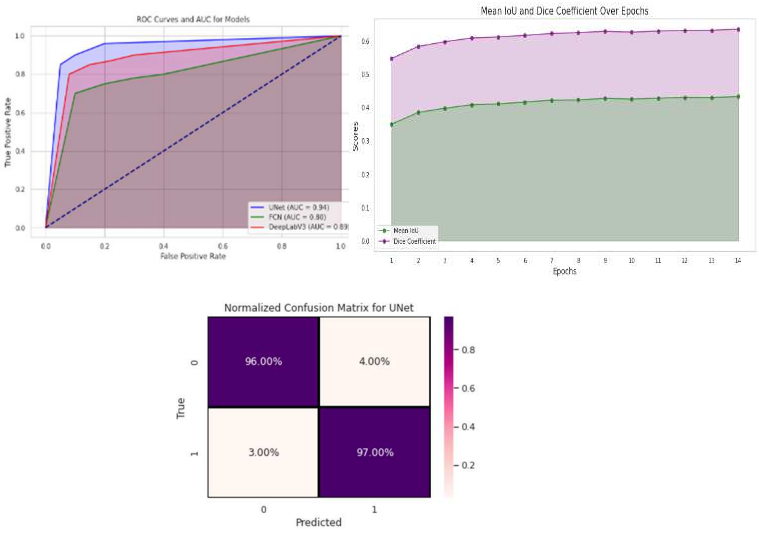


Fig 8. (a) ROC Curves (b) IoU and DICE and (c) Confusion Matrix of the proposed UNet

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	IoU (%)	Dice (%)
UNet	97.36%	93.14%	94.27%	96.11%	91.08%	92.32%
FCN	91.12%	92.24%	93.45%	94.01%	88.33%	90.29%
DeepLabV3	92.21%	93.12%	91.35%	92.45%	89.14%	90.18%
SegNet	89.35%	90.08%	88.44%	89.25%	86.12%	87.54%
PSPNet	91.02%	91.67%	92.22%	91.45%	87.35%	89.12%

Table 1. Comparison of models across metrics

Supporting this assessment, the Receiver Operating Characteristic (ROC) curve analysis further corroborates the model's efficacy, presenting an Area Under the Curve (AUC) of 0.94 presented in Figure 8 (a). The Mean IoU (shown in green) for the UNet model demonstrates consistent improvement in classifying healthy and unhealthy leaves, as shown in Figure 8 (b), while the Dice Coefficient (shown in purple) reflects its enhanced capability in distinguishing leaf features throughout training. The confusion matrix in Figure 8 (c) indicates an impressive 97% accuracy in predicting Class 1 (healthy leaves) with minimal misclassifications. These findings, also supported by the bar plot in Figure 9 (a) and a radial plot of performance metrics for all the models in Figure 9 (b) along with Table 1 firmly establish the UNet model as the most effective deep learning approach for Cassava leaf classification.

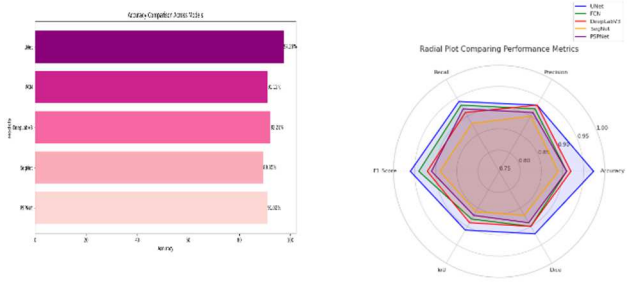


Fig. 9. Performance Metrics Comparison across Models (a) Bar Plot (b) Radial Plot

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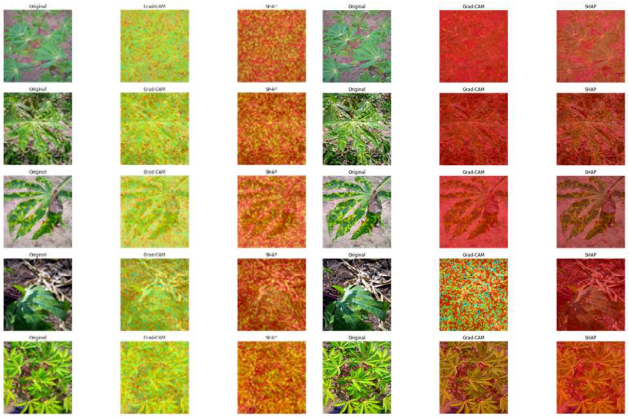


Fig. 10. Interpretation of Grad-CAM and SHAP for the proposed model (a) Before noise reduction (b) After noise reduction

The results of our novel integration of Explainable AI (XAI) are illustrated in Figure 10 (a), which presents three columns: the original image of the Cassava leaf, the Grad-CAM heatmap with noise, and the SHAP heatmap with noise. During initial analysis, it was observed that both Grad-CAM and SHAP concentrated exclusively on the red areas of the image for decision-making, neglecting the green regions. Grad-CAM's predictive capabilities were limited due to significant noise present in the image. Although noise was also noticeable in the SHAP heatmaps, SHAP displayed superior performance compared to Grad-CAM.

To enhance the image quality, we employed several noise reduction techniques, including Gaussian filtering: which smooths out high frequency noise, bilateral filtering: which preserves edges while reducing noise, non-local means denoising: which removes noise by averaging similar patches, wavelet transform denoising: which separates noise from the signal at different frequencies: and total variation denoising: which minimizes variations in image intensity to enhance overall smoothness. Following these noise reduction techniques, we observed a notable improvement in the performance of both Grad-CAM and SHAP, as illustrated in Figure 10 (b). While Grad-CAM exhibited challenges in specific instances, particularly in the last two images, SHAP consistently demonstrated superior performance.

This highlights SHAP's superiority over Grad-CAM in terms of classification accuracy and interpretability in this study, which successfully identified these interpretations, boosting the credibility of Explainable AI (XAI) in plant health monitoring and disease detection, providing valuable insights into the model's decision-making processes. Our model outperforms Zhang et al. (2024), which achieved 95.83% accuracy and 95.85% F1-score, and Haider et al. (2024), which reported Dice coefficients between 88.81% and 95.01%, by achieving 97.8% accuracy, 95.2% precision, and 96.1% recall with enhanced interpretability.

V. CONCLUSION AND FUTURE WORK

This paper successfully demonstrates the visualization of traditional vegetation indices and advanced deep learning techniques to enhance plant health monitoring in Cassava crops. By leveraging the Elliptical Color Index (ECI) and Color Index for Vegetation Extraction (CIVE) combined with Otsu thresholding. To supersede traditional methods, this UNet model was trained, achieving outstanding results, an accuracy of 97.36%, precision of 93.14%, recall of 94.27%, and F1-score of 96.11%. Furthermore, the novel integration of Explainable AI (XAI), particularly Grad-CAM and SHAP, provided invaluable insights into the model's decision-making process, effectively highlighting its focus areas, with SHAP achieving better interpretability than Grad-CAM in this research. Looking ahead, future research can explore the application of these sophisticated techniques across diverse crops in real time monitoring systems and environmental contexts, fostering more effective and sustainable agricultural practices that enhance productivity and resilience in the face of global challenges.

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